Table 4: Qualitative examples of natural text completions for English-French, and Country-Capital

English-French		
Prompt:	The word "daily" means	The word 'link' can be understood as a synonym for
GPT-J	every day	'connection' or 'relation'. The term 'link' is used in...
GPT-J+English-French FV	tous les jours	'lien', et le mot 'lien' peut être compris comme un synonyme...
Country-Capital		
Prompt:	When you think of Netherlands,	
GPT-J	you probably think of tulips, windmills, and cheese. But the Netherlands is also home to...	
GPT-J+Country-Capital FV	you think of Amsterdam. But there are many other cities in the Netherlands. Here are some...	

Table 5: A direct decoding of the function vector for each task.

Task t	**Tokens in the distribution** $D(v_t)$ **in order of decreasing probability**
Antonym	' lesser', ' counterpart', 'wrong', ' negate', ' destroy'
Capitalize	' Vanilla', ' Copy', ' Adapter', ' Actor', ' Container'
Country-Capital	' Moscow', ' Bangkok', ' Paris', ' London', ' Madrid'
English-French	' âĶĬ', ' masc', ' ç¥l', ' embr', ' è'
Present-Past	'received', 'changed', 'killed', 'answered', ' Changed'
Singular-Plural	'cards', 'stocks', ' helmets', ' items', ' phones'

3.2 The Decoded Vocabulary of Function Vectors

Several studies have gleaned insights about the states and parameters of transformers by viewing them in terms of their decoded vocabulary tokens (Nostalgebraist, 2020; Geva et al., 2021; 2022; Dar et al., 2023; Belrose et al., 2023). Therefore we ask: can we understand an FV by decoding v_t directly to a token probability distribution? Results are shown in Table 5, which lists the top five tokens in the decoded distribution $D(v_t)$ for each task (additional tasks in Appendix I).

A clear pattern emerges: for most tasks, the decoded tokens lie within the task's output space. The Singular-Plural function vector decodes to a distribution of plural nouns, and Present-Past decodes to past-tense verbs. However, that is not the case for all tasks: English-French decodes to nonsense tokens, and the Antonym task decodes to words that evoke the abstract idea of reversal.

Given these meaningful decodings, we then ask whether the token vocabulary is sufficient to recreate a working function vector. That is, we begin with the token distribution $Q_t = D(v_t)$, and determine whether a function vector can be reconstructed if we know the top words in Q_t. Denote by Q_{tk} the distribution that resamples Q_t while restricting to only the top k words. We perform an optimization to reconstruct a $\hat{v}_{tk}$ that matches the distribution Q_{tk} when decoded (where CE is cross-entropy loss):

$$\hat{v}_{tk} = \arg\min_{v} \mathrm{CE}(Q_{tk}, D(v)) \tag{6}$$

In Table 6, the performance of $\hat{v}_{tk}$ is evaluated when used as a function vector. We find that, while it is possible to partially recreate the functionality of an FV, good performance typically requires more

Table 6: Performance of FV v_t is compared to the reconstruction $\hat{v}_{t\,100}$ that matches the top 100 tokens, and $\hat{v}_{t\,\mathrm{all}}$ that matches all 50k tokens in $D(v_t)$. The KL divergence between the $D(\hat{v}_{tk})$ and Q_{tk} are shown for each reconstruction as KL_k. Lowest performers for each task in red.

	Accuracy on zero-shot prompts				
Task t	v_t	$\hat{v}_{t\,100}$	KL_{100}	$\hat{v}_{t\,\mathrm{all}}$	$\mathrm{KL}_{\mathrm{all}}$
Antonym	$48.2 \pm 2.0\%$	$4.8 \pm 2.0\%$	0.0033	$39.6 \pm 2.6\%$	0.0137
Capitalize	$70.5 \pm 2.4\%$	$5.7 \pm 2.2\%$	0.0001	$51.5 \pm 11.6\%$	0.0053
Country-Capital	$83.2 \pm 2.7\%$	$58.1 \pm 18.5\%$	0.0002	$29.0 \pm 15.1\%$	0.0019
English-French	$44.7 \pm 1.2\%$	$4.8 \pm 1.7\%$	0.0	$42.0 \pm 5.6\%$	0.0056
Present-Past	$19.7 \pm 5.9\%$	$4.4 \pm 1.4\%$	0.0052	$6.8 \pm 2.6\%$	0.0139
Singular-Plural	$47.0 \pm 3.4\%$	$23.3 \pm 6.1\%$	0.0	$27.4 \pm 4.7\%$	0.0145

than 100 vocabulary tokens. In other words, knowledge of the top decoded tokens of $D(v_t)$ is usually not enough on its own to construct a working function vector. That suggests that the FV contains some needed information beyond that expressed by its top decoded tokens.

3.3 Vector Algebra on Function Vectors

(a) Input: `"Italy, Russia, China, Japan, France"`

FV	Task	Expected Output
v_{AC}	First-Copy	`Italy`
v_{AD}	First-Capital	`Rome`
v_{BC}	Last-Copy	`France`
v^*_{BD}	Last-Capital	`Paris`

(b)

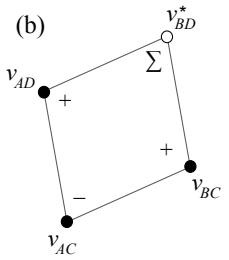

Figure 5: (a) A set of three list-oriented tasks that can be composed to a fourth task using FV vector algebra. (b) The parallelogram arrangement of the fourth vector v^*_{BD} when it is composed out of the other three FVs.

Although Table 6 suggests that function vectors cannot be understood as simple semantic vector offsets on word embeddings, we can ask whether function vectors obey semantic vector algebra over the more abstract space of *functional* behavior by testing the composition of simple functions into more complex ones. We begin with three conceptually decomposable ICL tasks: the list-oriented tasks First-Copy, First-Capital, and Last-Copy, as illustrated in Figure 5a. Using ICL, we collect FVs for all three tasks and denote them v_{AC}, v_{BC}, and v_{AD}.

Then we form a simple algebraic sum to create a new vector that we will denote v^*_{BD}.

$$v^*_{BD} = v_{AD} + v_{BC} - v_{AC} \tag{7}$$

$$\text{Last-Capital}^* = \text{Last-Copy} + \text{First-Capital} - \text{First-Copy} \tag{8}$$

In principle we could expect v^*_{BD} to serve as a new function vector for a new composed task (Last-Capital). We perform several similar task compositions on a variety of tasks. In each case, we combine a task with First-Copy and Last-Copy to produce a composed Last-$*$ vector; then, we test the accuracy of v^*_{BD} as a function vector. We compare to the accuracy of the FV extracted from ICL, as well as accuracy of the same model performing the task using ICL. Results for GPT-J are reported in Table 7; see Appendix K for results for Llama 2 (13 and 70 billion parameter models). We find that some FVs can be composed, with algebraic compositions outperforming FVs and even ICL on some tasks. Other tasks, including some for which ICL and FVs perform well, resist vector composition.

The ability to compose the tasks that we have demonstrated may hinge on the fact that "word-selection" from context and "word-transformation" are different components of language tasks that could involve FVs triggering complementary underlying mechanisms (e.g., one for locating and extracting input and another for transforming it). We therefore believe that FV composition may be a useful tool for further understanding the mechanisms of LMs.

Table 7: The accuracy of ICL, calculated FV v_{BD} zero-shot interventions, and vector-composed v^*_{BD} zero-shot interventions when performing several list-oriented tasks. Unlike our previous evaluations, here we measure performance on *all* available samples of the task, without restriction to the subset where the LM predicts correct output. In a few cases, composed function vector intervention v^*_{BD} can perform a task better than ICL.

Task	ICL (ten-shot)	v_{BD} (FV on zero-shot)	v^*_{BD} (sum on zero-shot)
Last-Antonym	0.25 ± 0.02	0.02 ± 0.01	0.07 ± 0.02
Last-Capitalize	0.91 ± 0.02	0.64 ± 0.03	0.76 ± 0.04
Last-Country-Capital	0.32 ± 0.02	0.15 ± 0.03	$\mathbf{0.60} \pm 0.02$
Last-English-French	0.45 ± 0.04	0.16 ± 0.02	0.06 ± 0.02
Last-Present-Past	0.89 ± 0.02	0.18 ± 0.02	0.29 ± 0.03
Last-Singular-Plural	0.90 ± 0.01	0.28 ± 0.01	0.29 ± 0.02
Last-Capitalize-First-Letter	0.75 ± 0.01	0.76 ± 0.02	$\mathbf{0.95} \pm 0.00$
Last-Product-Company	0.35 ± 0.03	0.30 ± 0.02	$\mathbf{0.41} \pm 0.03$

4 Related Work

A cousin to function vectors has been independently observed in concurrent work by Hendel et al. (2023); they study causal effects of $\mathbf{h}_\ell^t$ (similar to Section 2.1) on a different set of models and tasks.

Task Representations. Our work shows that it is possible to extract FVs with strong causal effects from LLMs; this is an advance over previous examinations that have *added* task representations to LLMs, e.g. Lampinen & McClelland (2020); Shao et al. (2023); Mu et al. (2023); Panigrahi et al. (2023); Ilharco et al. (2023), who devised ways to create compositional task encodings for LLMs using metamappings, codebooks, soft-prompts or sets of model parameter perturbations that Ilharco et al. calls *task vectors*. Unlike these previous works that create function representations, we find that compact FVs *already exist* within LLMs and show how to extract them. Likewise Lake & Baroni (2018); Hill et al. (2018) show RNN hidden states cluster on similar tasks. Our work differs because FVs are causal, not just correlative, so they can be explicitly extracted and inserted.

In-Context Learning. Since its observation in LLMs by Brown et al. (2020), ICL has been studied intensively from many perspectives. The role of ICL prompt forms has been studied by Reynolds & McDonell (2021); Min et al. (2022); Yoo et al. (2022). Models of inference-time metalearning that could explain ICL have been proposed by Akyürek et al. (2022); Dai et al. (2023); Von Oswald et al. (2023); Li et al. (2023b); Garg et al. (2022). Analyses of ICL as Bayesian task inference have been performed by Xie et al. (2021); Wang et al. (2023c); Wies et al. (2023); Hahn & Goyal (2023); Zhang et al. (2023); Han et al. (2023). And ICL robustness under scaling has been studied by Wei et al. (2023); Wang et al. (2023b); Pan et al. (2023). Our work differs from those studies of the externally observable behavior of ICL by instead focusing on mechanisms *within* transformers.

Mechanisms of task performance in LMs. Our work is related to Merullo et al. (2023); Halawi et al. (2023) which analyze components during execution of ICL tasks and identify causes of false statements. Also related are several methods that modify activations at inference time to steer LM behavior (Li et al., 2023a; Hernandez et al., 2023a; Subramani et al., 2022; Turner et al., 2023; Rimsky et al., 2023; Liu et al., 2023; Zou et al., 2023). Our work is consistent with Wang et al. (2023a) which observes salience of label tokens during ICL, Wang et al. (2022b) which observes individual neurons that correlate with specific task performance, and Variengien & Winsor (2023) shows task requests are processed in middle layers. We measure causal mediators across a distribution of different tasks to find a generic function-invocation mechanism that *identifies and distinguishes* between tasks.

Mechanistic Interpretability. We also build upon the analyses of Elhage et al. (2021) and Olsson et al. (2022), who observed prevalent in-context copying behavior related to jumps in performance during training. We isolate FVs using causal mediation analysis methods developed in Pearl (2001); Vig et al. (2020); Meng et al. (2022); Wang et al. (2022a); Geva et al. (2023). Our examination of FVs in vocabulary uses the logit lens of Nostalgebraist (2020); Geva et al. (2021); Dar et al. (2023).

Analyzing the Attention Mechanism. Our work is related to previous attention-weight analyses (Voita et al., 2018; Clark et al., 2019; Voita et al., 2019; Kovaleva et al., 2019; Reif et al., 2019; Lin et al., 2019; Htut et al., 2019; Kobayashi et al., 2020), that have found attention weights that align with linguistic structures. Our work is motivated by the observation that attention weights alone do not fully explain model outputs (Jain & Wallace, 2019; Wiegreffe & Pinter, 2019; Bibal et al., 2022). The focus of our paper is to extend our understanding of attention by investigating the *content* of the information transported by the attention heads in ICL to open a new window into the human-interpretable role that attention plays in language processing.

5 Discussion

Function vectors are a surprising finding. The metalearning capabilities of LLMs that have been studied since Brown et al. (2020) seem complex enough be inscrutable. Yet in this paper we have found a simple mechanism in a range of transformer LLMs that is common across tasks and robust to shifts in context: *function vectors* (FVs) that represent the task within a hidden state. FVs can be explicitly extracted from a small fixed set of attention heads that can be easily identified, and these FVs represent a range of tasks just as simply as word vectors (Mikolov et al., 2013)—yet our findings also reveal FVs must be a distinct phenomenon (Appendix A). Although FVs are not yet a complete accounting of how ICL works, they do provide new clarity on one level of mediation within ICL, and they open up a new path for future research to fully characterize function execution within LLMs.